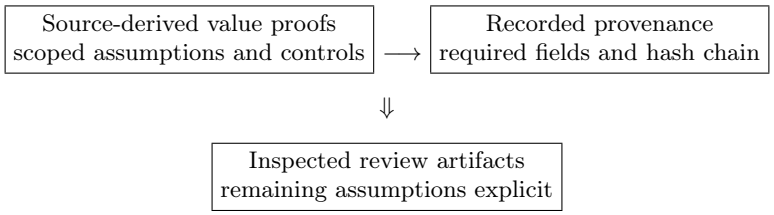
Weighted output, graph cycles and native comparator

This packet records three parallel formal components, with exact source snapshots and separate machine receipts. No scientific cohort was executed; the frozen 1,260-target scope and expired campaign limits remain unchanged.



Weighted CZero: 14 main declarations, seven controls and 21 axiom readbacks passed. For arbitrary nonzero input weights, the literal crossed coefficients and output payload decode with x/z^2 and y/(z^5 w). The theorem derives nonzero output weights and curve membership from the normalized curve and input divisibility premises. Raw reconstruction and branch-guard reflection are included. The controls retain an incorrect cubic-denominator counterexample.

Graph cycles: ten main declarations, eight controls and 18 axiom readbacks passed. Opposite-sign cycles give the native constraint equation; equal-sign cycles give an affine root anchor under a nonzero-two assumption. Existing-anchor differences, nonzero-coefficient extraction and zero-coefficient consistency are proved. Characteristic-two and sign controls are retained.

Native Kummer comparator: eight main declarations, ten controls and 18 axiom readbacks passed. The actual narrow UInt64 multiplication is used. Canonical coordinates and prime p <= 2^32 give a product below 2^64 and make the sixteen Boolean cross-products agree with field proportionality. The zero tests agree under canonicality. Outside-bound wraparound and noncanonical-zero counterexamples are retained.

Independent source reviews and fresh replay passed at each stated scope: weighted output 27 readbacks, graph cycles 26, native comparator 24. The reviews include twenty additional controls in total. One reviewer expected-value error was corrected and its failed receipt retained. Producer modules are archived without rewriting their receipts. Review sessions were assigned against stable private snapshots before this publication; this is a sequencing deviation from the publish-before-dispatch rule, not a claim of a precommitted review plan. Final reviews must bind the same source bytes.

Remaining interfaces are concrete: field-value output is not yet a divisor-class addition proof; cycle algebra does not establish the mutable forest invariant or native inversion; comparator correctness requires caller bounds and canonicality and does not establish geometric fibers or transition closure. No timing or cost ratio is inferred from these proofs.

Reports, editable SVGs, status JSON, native source excerpts, failed attempts and axiom readbacks are retained per component. The shared session-log path remains separately reserved; this packet carries its own journal append pending authorization.

Current-main integration was attempted at c3a92a2feb2bd8d5ce503cd8f1b7cfe7e0ba4978 and aborted after a shared session-log conflict; no evidence record was resolved by editing. Review workflow completeness remains false because publication preceded neither dispatch nor a precommitted plan. Technical checks and workflow status are reported separately.